

MedTrack DV

HospitalOperations & Patient Analytics Dashboard

Project Documentation

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1. Dataset Sources

The MedTrack DV dashboard is built on a single patient-encounter dataset loaded into the Power BI data model. Each row represents one patient admission record, enriched with department-, staff-, bed-, and equipment-level attributes used to derive operational KPIs.

1.1 Data Grain

Patient ID is the primary key / grain of the fact table — one row per patient admission.

1.2 Fields Used in the Report

Field	Type	Description
Patient ID	Text/Key	Unique identifier for each patient record; the row-level grain of the dataset.
Admission Month	Date	Calendar month of patient admission; drives monthly trend visuals.
Admission Quarter	Date/Category	Quarter of admission; used as a global slicer across all report pages. Hospital
Department	Text/Category	department the patient was admitted to (10 distinct departments). Patient
Gender	Category	gender, used for demographic breakdowns.
Hospital Type	Category	Classification of the hospital/facility (e.g., Government, Private). Geographic
State	Category	state associated with the admission/facility.
Length of Stay	Numeric	Number of days the patient remained admitted.
Billing Amount	Numeric (Currency)	Total billed amount associated with the patient encounter.
Transferred	Boolean/Flag	Indicates whether the patient was transferred between departments/facilities.
Equipment Status	Category	Operational status of medical equipment (e.g., Available, In Use, Under Maintenance).
Equipment_Usage_Duration_Hours	Numeric	Number of staff allocated per department.
Staff Count	Numeric	Derived percentage of staff capacity utilised, by department.
Staff_Utilization_%_Derived	Calculated	Count of occupied beds, used in bed occupancy and treatment capacity
Beds_Occupied_Count	Numeric	Derived total bed capacity per department.
Dept_Bed_Capacity_Derived	Calculated	Derived percentage of beds occupied relative to capacity.
Bed_Occupancy_Rate_%	Calculated	

1.3 Derived / Calculated Fields

Several fields ending in "_Derived" or "%" (e.g., Staff_Utilization_%_Derived, Dept_Bed_Capacity_Derived, Bed_Occupancy_Rate_%) are calculated within the data model to support department-level rollups that are not present directly in the source extract. Review these periodically against source system definitions to ensure they remain aligned with hospital reporting standards.

1.4 Data Refresh

Confirm with the data/BI team whether the underlying dataset is refreshed on a scheduled basis (e.g., daily/weekly) or is a static extract, and record the refresh cadence and data-as-of date here once confirmed.

2. KPI Definitions

The table below documents every KPI surfaced on the four dashboard pages, along with its current reported value (per the latest QA validation) and its calculation logic.

KPI	Reported Value	Definition / Formula
Total Admissions	10,00	Count of distinct Patient ID records in the reporting period.
Overall Occupancy Rate	0	Beds Occupied ÷ Total Bed Capacity, aggregated across all departments.
Overall Readmission Rate	36.8%	Count of admissions flagged as readmissions ÷ Total Admissions.
Average Length of Stay	49.9%	Average of Length of Stay across all admitted patients (days).
Transfer Rate	78.78	Count of patients where Transferred = Yes ÷ Total Admissions.
Number of Departments	17.6%	Distinct count of Department values in the dataset.
Department Efficiency Score	10	Composite index combining throughput, average LOS, and resource utilisation by department.
Equipment Availability Rate	29.15 32.9%	Count of equipment with Equipment Status = Available ÷ Total Equipment Records.
Staff Utilization Rate	70.8%	Average of Staff_Utilization_%_Derived, weighted by Staff Count across departments.
Bed Occupancy Rate	37.3	Beds_Occupied_Count ÷ Dept_Bed_Capacity_Derived, aggregated across departments.
Total Patient Transfers	%	Count of patient records where Transferred = Yes.
Total Discharges	1,760	Count of patient records with a recorded discharge date/event.
Patient Volume (by dept.)	—	Count of Patient ID records, split by Department.

2.1 Notes on Interpretation

- All rate/percentage KPIs are expected to fall within 0–100%; any value outside this range indicates a data quality issue (see QA-11).
- Readmission Rate and Transfer Rate depend on accurate flagging in the source system (readmission flag, Transferred field) — see QA-13 and QA-14.
- Department Efficiency Score is a composite/index metric; confirm its exact weighting formula with the report author if it needs to be reproduced outside Power BI.

3. Dashboard Guide

MedTrack DV consists of four connected report pages, navigable via on-page buttons (top-right of each page). All pages share a consistent visual language: KPI cards at the top, trend/comparison charts below, and global slicers for filtering.

3.1 Global Filters & Slicers

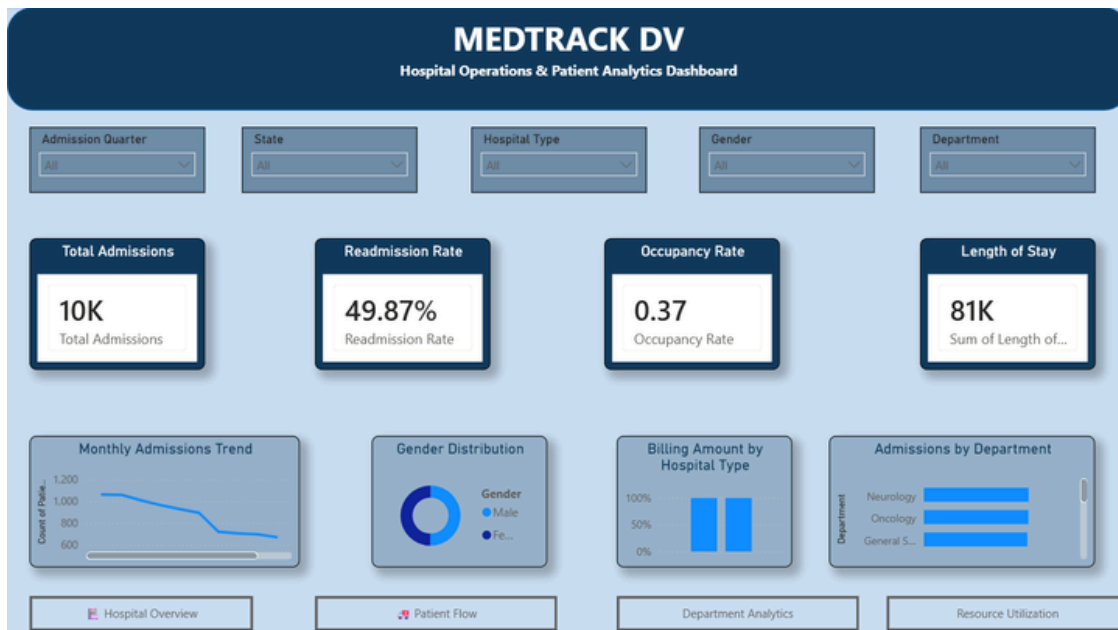
Slicer / Filter	Available On
State	Hospital Overview
Hospital Type	All 4 pages
Gender	All 4 pages
Department	All 4 pages
Admission Quarter	All 4 pages

3.2 Navigation

Each page includes action buttons that link directly to the other three pages, allowing users to move between the Hospital Overview, Patient Flow, Department Analytics, and Resource Utilisation views without using the page tabs.

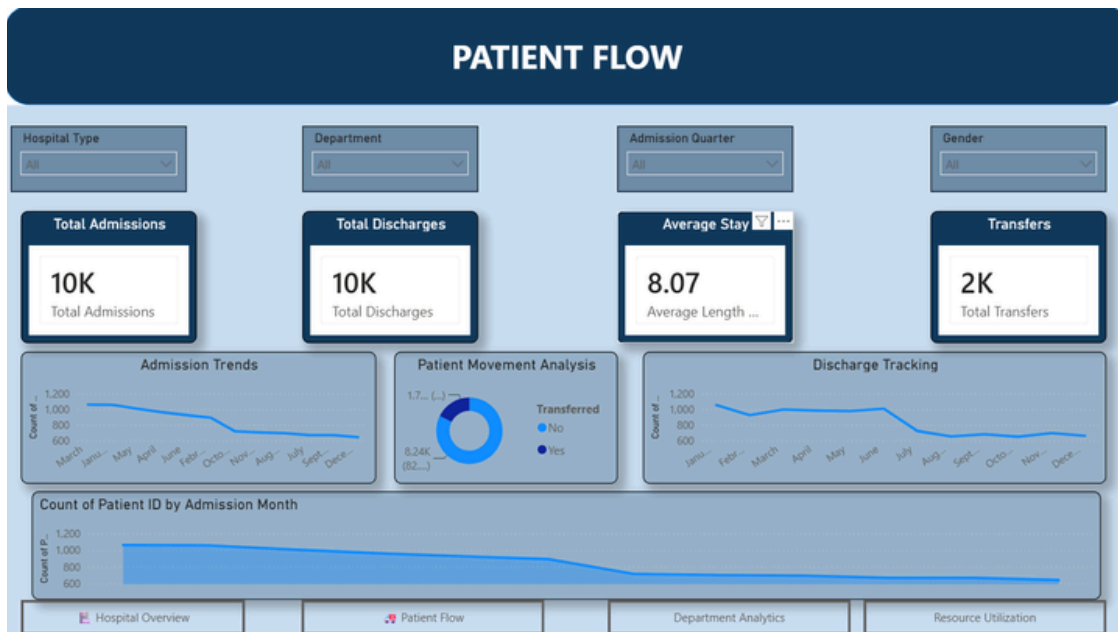
3.3 Page1—Hospital Overview

Executive summary of hospital-wide performance for leadership review. KPI cards: Total Admissions, Readmission Rate, Occupancy Rate, Length of Stay. Charts: Monthly Admissions Trend, Gender Distribution, Admissions by Department, Billing Amount by Hospital Type.



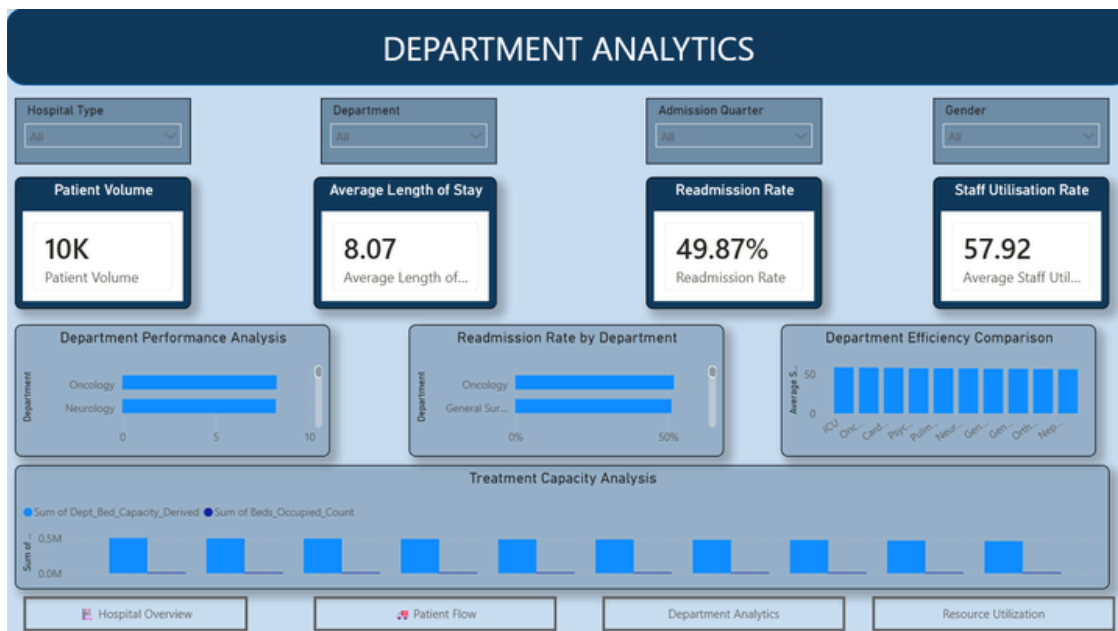
3.4 Page 2 — Patient Flow

Tracks patient movement through the hospital, from admission to discharge, including transfers. KPI cards: Total Admissions, Total Discharges, Average Stay, Transfers. Charts: Admission Trends, Discharge Tracking, Patient Movement Analysis, and a patient-flow area chart.



3.5 Page 3 — Department Analytics

Compares performance and capacity across the 10 hospital departments. KPI cards: Patient Volume, Average Length of Stay, Readmission Rate, Staff Utilisation Rate. Charts: Department Performance Analysis, Readmission Rate by Department, Department Efficiency Comparison, Treatment Capacity Analysis.



3.6 Page 4 — Resource Utilisation

Monitors operational resources —staff, beds, and equipment — to support capacity planning. KPI cards: Total Transfers, Staff Utilisation Rate, Equipment Utilisation, Bed Occupancy Rate. Charts: Resource Availability Analysis, Equipment Utilization Tracking, Staff Allocation by Department, Bed Utilization by Department, Capacity Planning Insights.



4. Healthcare Operations Methodology

This section outlines the operational logic and assumptions underpinning the dashboard's metrics, to ensure consistent interpretation across stakeholders.

4.1 Patient Flow Model

The dashboard models the patient journey in three stages — Admission, In-Care (including any Transfers between departments), and Discharge. Length of Stay is measured as the duration between admission and discharge; Transfer Rate captures the proportion of patients who moved between departments or facilities during their episode of care.

4.2 Capacity & Resource Management

Bed Occupancy Rate and Treatment Capacity Analysis compare Beds Occupied against derived Department Bed Capacity to flag over- or under-utilised departments. Staff Utilization Rate and Equipment Availability Rate apply the same occupied-vs-capacity logic to workforce and equipment assets, supporting proactive resource allocation.

4.3 Quality & Risk Indicators

Readmission Rate acts as a proxy quality indicator — elevated rates by department can highlight potential gaps in discharge planning or care quality that warrant clinical review. Department Efficiency Score combines throughput and resource metrics into a single comparative index for department-level benchmarking.

4.4 Governance & Validation

All KPIs are validated against source records through the accompanying QA Checklist (MedTrack_DV_QA_Checklist.docx), which verifies KPI accuracy, healthcare-metric integrity (no invalid percentages or negative values), patient-flow consistency, and dashboard interactivity across all four report pages.